where it is assumed that $\Omega \gg 1$. The differential equations describing this system are

$$\dot{q} = p,$$
$$\dot{p} = -(1 + \Omega^2)q.$$

The frequency is therefore $\bar{\Omega} = \sqrt{1 + \Omega^2}$. In the limit of small inner step, we solve the system with Hamiltonian $H_{\mathrm{F}} = p^2/2 + \Omega^2 q^2/2$ *exactly*, meanwhile punctuating with impulses from U_{S}, thus the MTS flow map approximation, which is linear in this case, can be written in the form of a matrix-vector multiply

$$\begin{bmatrix} Q \\ P \end{bmatrix} = \boldsymbol{W}_h \begin{bmatrix} q \\ p \end{bmatrix},$$

where

$$\boldsymbol{W}_h = \boldsymbol{S}_{h/2} \boldsymbol{F}_h \boldsymbol{S}_{h/2},$$

with

$$\boldsymbol{S}_h = \begin{bmatrix} 1 & 0 \\ -h & 1 \end{bmatrix}, \qquad \boldsymbol{F}_h = \begin{bmatrix} \cos(\Omega h) & \Omega^{-1} \sin(\Omega h) \\ -\Omega \sin(\Omega h) & \cos(\Omega h) \end{bmatrix}.$$

Both of these matrices have unit determinant for any h, so the product of the eigenvalues of $\boldsymbol{W}_h$ is 1. On the other hand, a short calculation shows that

$$\lambda_1 + \lambda_2 = \mathrm{Tr}(\boldsymbol{W}_h) = 2\cos(h\Omega) - \frac{h}{\Omega} \sin(h\Omega).$$

For a given h, the eigenvalues of $\boldsymbol{W}_h$ are either complex conjugates or they are real and reciprocal to one another. If they are complex conjugates with nonzero imaginary part, then we have $\lambda_1 \lambda_2 = \lambda_1 \bar{\lambda}_1 = 1$, so $|\lambda_1| = 1$ and the two eigenvalues are clearly distinct and on the unit circle in the complex plane.

If we consider a relatively small step of size $h\Omega = k\pi$ we find that the eigenvalues cross, and the system is unstable. For example, at $h = \pi/\Omega$, we have

$$\boldsymbol{F}_h = -\boldsymbol{I} \Rightarrow \boldsymbol{W}_h = -\boldsymbol{S}_{h/2}^2 = -\begin{bmatrix} 1 & 0 \\ -h & 1 \end{bmatrix},$$

so that

$$\boldsymbol{W}_h^n = (-1)^n \begin{bmatrix} 1 & 0 \\ -nh & 1 \end{bmatrix},$$